



M-Tech Production & Marketing

Technology | Training | Marketing | Projects

TASK 7 REPORT

Internship Program — Batch 2026

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Date:	11 July 2026

1. Introduction

This report presents the work completed for **Task 07** of the M-Tech Production & Marketing AI/ML Internship Program, Batch 2026. The internship focuses on practical exposure to core concepts in Artificial Intelligence and Machine Learning, equipping interns with skills applicable to real-world technology and marketing challenges.

M-Tech Production & Marketing is a forward-looking company based in Haripur, KPK, specializing in Technology, Training, Marketing, and Projects. The internship is structured in a 9-Phase roadmap designed to progressively build theoretical knowledge and hands-on project experience.

2. Objectives

- To understand the importance of model evaluation in Machine Learning.
- To learn how model performance is measured using different evaluation metrics.
- To study the concepts of training, validation, and testing datasets.
- To understand overfitting and underfitting and how to avoid them.
- To learn different model validation techniques, including Cross-Validation.
- To build a strong foundation for selecting and improving machine learning models.

3. Task Description

In this experiment, I studied the process of Model Evaluation and Validation in Machine Learning. I learned how trained models are assessed using different performance metrics depending on whether the task is regression or classification. The experiment covered concepts such as Training Set, Validation Set, Test Set, Accuracy, Precision,

Recall, F1-Score, Confusion Matrix, Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Cross-Validation. I also explored how proper validation helps prevent overfitting and ensures that models perform well on unseen data.

4. Work Done

During this experiment, I completed the following activities:

- Studied the importance of model evaluation in Machine Learning.
- Learned the difference between training, validation, and testing datasets.
- Explored common evaluation metrics for Classification Models, including:
 - Accuracy
 - Precision
 - Recall
 - F1-Score
 - Confusion Matrix
- Studied evaluation metrics for Regression Models, including:
 - Mean Absolute Error (MAE)
 - Mean Squared Error (MSE)
 - Root Mean Squared Error (RMSE)
 - R^2 Score
- Learned the concepts of Overfitting and Underfitting.
- Explored techniques to improve model performance through proper validation.
- Studied K-Fold Cross-Validation as a method for evaluating model reliability.

5. Challenges Faced

- Understanding when to use different evaluation metrics for regression and classification models.
- Interpreting confusion matrix results correctly.
- Differentiating between validation and testing datasets.
- Understanding how Cross-Validation improves model reliability and reduces bias.

6. Key Learnings

Through this experiment, I learned:

- The importance of evaluating machine learning models before deployment.
- How different evaluation metrics measure different aspects of model performance.
- The role of training, validation, and testing datasets in developing reliable models.
- How to identify overfitting and underfitting problems.
- The importance of Cross-Validation for improving model generalization.
- Best practices for selecting suitable evaluation techniques based on the machine learning problem.

7. Conclusion

This experiment provided a strong understanding of Model Evaluation and Validation, which are essential steps in the machine learning workflow. I learned how to measure model performance using various evaluation metrics, validate models using appropriate techniques, and identify issues such as overfitting and underfitting. These concepts are critical for developing accurate, reliable, and robust machine learning models that perform effectively on real-world data.

Intern Signature: Hamza Badshah**Supervisor:** _____**Name:** Hamza Badshah**Name:****Date:** 11 July 2026**Designation:**